Lecture notes: Mathematical Optimization and Applications

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CHAPTER 1

Linear programming

The aim of this chapter is to introduce linear programs and their duality theory. We start with the convex sets that appear as feasible sets of linear programs, and with Farkas Lemma, which characterizes when a system of linear inequalities has no solution. Parts of this chapter and of Chapter 2 follow [**12**, Sec. 2.2–2.3]. The chapter contains:

i) notation for vectors, matrices, and the componentwise order (Section 1.1);
ii) convex sets, cones, convex hulls, and polyhedra (Section 1.2);
iii) Farkas Lemma, with the proof of the harder direction in Appendix A (Section 1.3);
iv) primal and dual linear programs, feasibility and optimal values, and how to bring a linear program into standard form (Section 1.4);
v) weak duality, the duality theorem, and complementary slackness (Section 1.5);
vi) infeasibility certificates (Section 1.6).

1.1. Notation

Throughout, $A \in \mathbb{R}^{m\times n}$ is a real matrix with rows $a_1^T, \dots, a_m^T$,

$$A = \begin{pmatrix} \cdots a_1^T \cdots \\ \vdots \\ \cdots a_m^T \cdots \end{pmatrix}, \tag{1}$$

so that $a_i \in \mathbb{R}^n$. The vectors $b = (b_1, \dots, b_m)^T \in \mathbb{R}^m$ and $c \in \mathbb{R}^n$ are column vectors. The zero vector is $\mathbf{0}$, the all-ones vector is $\mathbf{1}$, and I is the identity matrix. For $r, s \in \mathbb{R}^m$ we write $r \le s$ if $r_i \le s_i$ for all $i = 1, \dots, m$ (componentwise order). In particular, $r \ge 0$ means $r_i \ge 0$ for all i. The set of vectors $x \in \mathbb{R}^n$ with $x \ge 0$ is the non-negative orthant $\mathbb{R}^n_+$. The expression $Ax \le b$ denotes a system of m linear inequalities $a_i^T x \le b_i$, $i = 1, \dots, m$.

We will use the following property of the componentwise order several times:

$$p \ge 0 \quad \text{and} \quad r \le s \quad \Longrightarrow \quad p^T r \le p^T s\,. \tag{2}$$

Indeed, if $r_i \le s_i$ and $p_i \ge 0$, then $p_i r_i \le p_i s_i$. Summing over all i gives $p^T r = \sum_i p_i r_i \le \sum_i p_i s_i = p^T s$. In words: a non-negative combination of valid linear inequalities is a valid linear inequality.

1.2. Convex sets and polyhedra

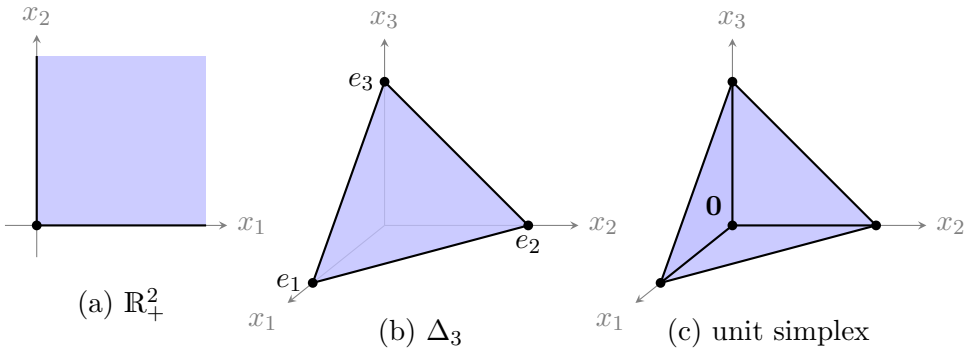
The aim of this section is to introduce the convex sets that appear as feasible sets of linear programs.

1.2.1. Cones. A subset K of a vector space is a cone if $\lambda x \in K$ for all $x \in K$ and $\lambda \ge 0$. See Figure 2(a).

1.2.2. Convex sets. A subset C of a vector space is convex if $\lambda x + (1-\lambda)y \in C$ for all $x, y \in C$ and $\lambda \in [0, 1]$, i.e. if it contains the segment between any two of its points. A convex cone is a cone that is convex. Equivalently, $\lambda x + \mu y \in K$ for all $x, y \in K$ and $\lambda, \mu \ge 0$.

1.2.3. Convex hull. A convex combination of $x_1, \dots, x_k$ is a sum $\sum_i \lambda_i x_i$ with $\lambda_i \ge 0$ and $\sum_i \lambda_i = 1$. The convex hull $\operatorname{conv}(S)$ of a set S is the set of all convex combinations of finitely many points of S. It is the smallest convex set containing S. See Figure 2(c), (d).

1.2.4. Polyhedra. A *halfspace* is a set $\{x \in \mathbb{R}^n : a^T x \leq \gamma\}$ with $a \in \mathbb{R}^n$, $a \neq \mathbf{0}$, and $\gamma \in \mathbb{R}$. A *polyhedron* is an intersection of finitely many halfspaces, i.e. a set of the form $\{x \in \mathbb{R}^n : Ax \leq b\}$. A *polytope* is a bounded polyhedron. Every halfspace is convex, and an intersection of convex sets is convex, so every polyhedron is convex. A set is a polytope if and only if it is the convex hull of finitely many points (Minkowski–Weyl theorem, see [17, Ch. 1]).



(a) $\mathbb{R}^2_+$ (b) Δ_3 (c) unit simplex

FIGURE 1. The polyhedra of Example 1 for $n = 2$ and $n = 3$. (a) The non-negative orthant $\mathbb{R}^2_+$ is unbounded, so it is not a polytope. (b) The probability simplex $\Delta_3 = \operatorname{conv}\{e_1, e_2, e_3\}$ is a triangle in $\mathbb{R}^3$. (c) The unit simplex $\operatorname{conv}\{\mathbf{0}, e_1, e_2, e_3\}$ is a tetrahedron.

Example 1. Some convex sets, see Fig. 1.

i) The non-negative orthant $\mathbb{R}^n_+ = \{x : -x \leq \mathbf{0}\}$ is a polyhedron, but not a polytope.

ii) The probability simplex $\Delta_n = \{x \in \mathbb{R}^n : x \geq 0,\ \mathbf{1}^T x = 1\} = \operatorname{conv}\{e_1, \ldots, e_n\}$ is a polytope.

iii) The unit simplex $\{x \in \mathbb{R}^n : x \geq 0,\ \mathbf{1}^T x \leq 1\} = \operatorname{conv}\{\mathbf{0}, e_1, \ldots, e_n\}$ is a polytope.

◊

A *vertex* of a polyhedron Q is a point $v \in Q$ that is not of the form $v = \lambda x + (1-\lambda)z$ with $x, z \in Q$, $x \neq z$, and $\lambda \in (0,1)$.

Fact 2 ([13, Ch. 4], [15, Ch. 8])**.** *Let Q be a non-empty polyhedron that contains no line, for example a polytope or a polyhedron contained in $\mathbb{R}^n_+$. Then Q has a vertex. If a linear objective function has a finite optimal value over Q, then it has an optimal solution that is a vertex of Q.*

To summarize this section: polyhedra are the solution sets of finitely many linear inequalities; they are convex, and the bounded ones (polytopes) are the convex hulls of finitely many points.

1.3. Farkas Lemma

The aim of Farkas Lemma is to completely characterize when a system of linear inequalities $Ax \leq b$ has no solution.

THEOREM 3 (Farkas Lemma, [12, Lemma 2.6]). *Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Exactly one of the following two statements is true:*

i) There exists $x \in \mathbb{R}^n$ such that $Ax \leq b$.

ii) There exist $\lambda_1, \ldots, \lambda_m \geq 0$ such that

$$\sum_{i=1}^m \lambda_i a_i = \mathbf{0}\,,$$

$$\sum_{i=1}^m \lambda_i b_i = -1\,. \tag{3}$$

Condition ii) can be written in compressed form as $\lambda \in \mathbb{R}^m$, $\lambda \geq 0$ with $A^T\lambda = \mathbf{0}$ and $b^T\lambda = -1$.

Example 4. The system $x \le 1$, $-x \le -2$ in one variable $x \in \mathbb{R}$ has no solution, since it asks for $x \le 1$ and $x \ge 2$. Here $a_1 = 1$, $b_1 = 1$, $a_2 = -1$, $b_2 = -2$. The vector $\lambda = (1,1)^T$ satisfies statement ii): $\lambda_1 a_1 + \lambda_2 a_2 = 1 - 1 = 0$ and $\lambda_1 b_1 + \lambda_2 b_2 = 1 - 2 = -1$. Adding the two inequalities gives $0 \le -1$. ◊

This is due to Farkas [4], who stated it in an equivalent form. The version above and other variants are in [13, Ch. 6].

Statement ii) says: the non-negative combination of the inequalities $a_i^T x \le b_i$ with weights λ_i yields the inequality $0 \le -1$, which is a contradiction. Thus no $x \in \mathbb{R}^n$ satisfying $Ax \le b$ exists. We can think of λ as a short proof that i) fails [12, Sec. 2.2]. Farkas lemma says that such a proof (ii) exists whenever i) fails, and vice versa.

Proof that ii) excludes i). Suppose λ satisfies the constraints (3) and x satisfies $Ax \le b$. By (2) with $p = \lambda \ge 0$,

$$\lambda^T A x \le \lambda^T b\,. \tag{4}$$

But the left-hand side equals $(A^T\lambda)^T x = \mathbf{0}^T x = 0$, whereas the right-hand side equals $\lambda^T b = -1$. This is a contradiction. □

The other direction, that ii) holds whenever i) fails, is harder. The proof of this direction uses Fourier–Motzkin elimination and is given in Appendix A. It will not appear in the exam.

To summarize this section: the system $Ax \le b$ has no solution if and only if some non-negative combination of its inequalities yields the inequality $0 \le -1$.

1.4. Linear programming

1.4.1. Primal and dual LPs. The aim of this subsection is to define the primal and dual linear programs that we use throughout.

primal:

$$\begin{aligned} \alpha = \max \quad & c^T x \\ \text{subject to} \quad & Ax \le b \end{aligned} \tag{P}$$

Here the matrix $A \in \mathbb{R}^{m\times n}$ and the vectors $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ are the parameters, and $x \in \mathbb{R}^n$ is the vector of variables. In components, the constraints are $a_i^T x \le b_i$ for $i = 1, \dots, m$.

To every primal linear program one associates a dual program with the same parameters A, b, c [12, Sec. 2.2].

dual:

$$\begin{aligned} \beta = \min \quad & b^T y \\ \text{subject to} \quad & A^T y = c \\ & y \ge 0 \end{aligned} \tag{D}$$

Here $y \in \mathbb{R}^m$ is the vector of variables. The numbers α and β are the optimal values of (P) and (D); they are defined precisely in Section 1.4.2.

Note that the primal program has n variables and m inequality constraints. In contrast, the dual has m variables (one for each primal constraint), n equality constraints (one for each primal variable), and m sign constraints $y_i \ge 0$.

The objective value of every feasible point of the dual is an upper bound on the objective value of every feasible point of the primal. This is weak duality, proven in Section 1.5.

1.4.2. Feasibility and optimal values. The aim of this subsection is to fix the terms used to describe the outcome of a linear program.

Definition 5. *For the primal program* (P)*:*

i) A feasible point *(or feasible solution) is a vector $x \in \mathbb{R}^n$ with $Ax \le b$.*

ii) The feasible set *is $\mathcal{F}_P = \{x \in \mathbb{R}^n : Ax \le b\}$. $\mathcal{F}_P$ is a polyhedron.*

iii) The program is feasible *if $\mathcal{F}_P \neq \emptyset$, and* infeasible *otherwise.*

iv) The optimal value *is $\alpha = \sup\{c^T x : x \in \mathcal{F}_P\}$, with the convention $\alpha = -\infty$ if the program is infeasible.*

v) The program is unbounded *if $\alpha = +\infty$, i.e. if for every $M \in \mathbb{R}$ there is a feasible x with $c^T x > M$.*

vi) An optimal solution *is a feasible point x^* with $c^T x^* = \alpha$.*

For the dual program (D), the feasible set is $\mathcal{F}_D = \{y \in \mathbb{R}^m : A^T y = c,\ y \geq 0\}$ (a polyhedron), the optimal value is $\beta = \inf\{b^T y : y \in \mathcal{F}_D\}$ with the convention $\beta = +\infty$ if the program is infeasible, the program is unbounded if $\beta = -\infty$, and an optimal solution is a feasible point y^* with $b^T y^* = \beta$.

1.4.3. Transforming linear programs into standard form. The aim of this subsection is to bring any linear program into the primal standard form (P). Its dual is then (D).

There are many different conventions for linear programs (and also semidefinite programs), using min or max, constraints of type $\leq$, $\geq$, $=$, and sign constraints on the variables. These can be converted into each other [**12**, Sec. 2.2]. Instead of memorizing a dual for each convention, we convert to (P) and dualize from there. Four rules suffice:

i) Turn minimization into maximization: $\min\{c^T x : x \in \mathcal{F}\} = -\max\{(-c)^T x : x \in \mathcal{F}\}$, and both programs have the same optimal solutions.

ii) Reverse inequalities by $a^T x \geq \gamma \iff -a^T x \leq -\gamma$.

iii) Model equalities as pairs of inequalities:

$$Ax = b \quad \Longleftrightarrow \quad Ax \leq b \quad \text{and} \quad -Ax \leq -b\,. \tag{5}$$

iv) Sign constraints as inequalities: $x \geq 0 \iff -Ix \leq \mathbf{0}$.

In (P) the variables are not restricted in sign, so no further rules are needed.

Two further transformations go in the other direction, towards the form of (D) with equality constraints and non-negative variables. Solvers often expect one of these forms.

i) Inequalities by equalities with a slack variable z:

$$a^T x \leq \gamma \quad \Longleftrightarrow \quad \text{there is } z \geq 0 \text{ with } a^T x + z = \gamma\,. \tag{6}$$

In matrix form, $Ax \leq b$ if and only if there is $z \in \mathbb{R}^m$, $z \geq 0$, with

$$\begin{pmatrix} A & I \end{pmatrix} \cdot \begin{pmatrix} x \\ z \end{pmatrix} = b\,. \tag{7}$$

ii) Free variables by two non-negative variables: every $x \in \mathbb{R}^n$ can be written as

$$x = x_+ - x_- \quad \text{with} \quad x_+, x_- \in \mathbb{R}^n_+\,. \tag{8}$$

So inequality constraints can be exchanged for equality constraints together with additional non-negative variables, and free variables can be exchanged for twice as many non-negative variables.

Example 6. Consider

$$\begin{aligned} \min \quad & x_1 + x_2 + x_3 \\ \text{subject to} \quad & x_1 - x_2 = 0 \\ & x_1 + x_3 \leq 1\,. \end{aligned} \tag{9}$$

By rule i), its optimal value is $-\alpha$, where $\alpha = \max\{-x_1 - x_2 - x_3 : x_1 - x_2 = 0,\ x_1 + x_3 \leq 1\}$.

With rule iii) we can rewrite the constraints as $x_1 - x_2 \leq 0$, $-x_1 + x_2 \leq 0$, $x_1 + x_3 \leq 1$. This yields a primal program in standard form with

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad c = \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}. \tag{10}$$

The dual (D) is

$$\begin{aligned} \min \quad & y_3 \\ \text{subject to} \quad & y_1 - y_2 + y_3 = -1 \\ & -y_1 + y_2 = -1 \\ & y_3 = -1 \\ & y \geq 0\,. \end{aligned} \tag{11}$$

Note that the constraint $y_3 = -1$ contradicts $y_3 \geq 0$, and so the dual is infeasible. By the duality theorem (Section 1.5.2), the primal has no finite optimal value, so it is unbounded or infeasible. It is feasible ($x = \mathbf{0}$), so it is unbounded: the points $x = (-t, -t, 0)$ with $t \geq 0$ are feasible with $-x_1 - x_2 - x_3 = 2t \to \infty$. The original minimization problem has optimal value $-\infty$. ◊

1.5. Duality theory

The aim of this section is to show that dual objective values are upper bounds on primal objective values (weak duality), to state when the optimal values are equal (duality theorem), and to characterize optimal solutions (complementary slackness).

1.5.1. Weak duality.

Proposition 7 (Weak duality)**.** *For every pair of feasible solutions (x, y) of the primal and dual program, it holds that*

$$c^T x \leq b^T y\,. \tag{12}$$

Proof. Use the primal and dual constraints and (2) with $p = y \geq 0$ and $Ax \leq b$:

$$c^T x = (A^T y)^T x = y^T A x \leq y^T b = b^T y\,. \tag{13}$$

□

i) Taking the supremum over x and the infimum over y gives $\alpha \leq \beta$. With the conventions of Section 1.4.2, this also holds if one of the programs is infeasible.

ii) For any pair of feasible solutions (x, y) and optimal solutions (x^*, y^*),

$$c^T x \leq c^T x^* = \alpha \leq \beta = b^T y^* \leq b^T y\,. \tag{14}$$

So the objective value of a feasible x is a lower bound on α, and the objective value of a feasible y is an upper bound on α and on β.

iii) If $c^T x = b^T y$ for a feasible pair (x, y), then x and y are optimal solutions.

iv) If the primal is unbounded, $\alpha = +\infty$, then $\beta = +\infty$, so the dual is infeasible. If the dual is unbounded, the primal is infeasible.

v) The *duality gap* is $\Delta = \beta - \alpha \geq 0$.

vi) *Strong duality* holds if $\Delta = 0$, i.e. $\alpha = \beta$.

See Fig. 2.

1.5.2. Duality theorem. We now state the duality theorem and list which combinations of primal and dual outcomes can occur.

Theorem 8 (Duality theorem, [**12**, Thm. 2.8], [**13**, Ch. 6]). *If either the primal or the dual program has a finite optimal value, then both programs have optimal solutions, and their optimal values are equal, $\alpha = \beta$.*

We will not discuss the proof here. It can be derived from Farkas Lemma, see [**13**, Ch. 6]. Together with weak duality, the duality theorem gives two facts about a primal-dual pair:

i) If the primal or the dual has a finite optimal value, then so does the other, both are attained, and they are equal. (Duality theorem.)

ii) If the primal or the dual is unbounded, then the other is infeasible. (Weak duality.)

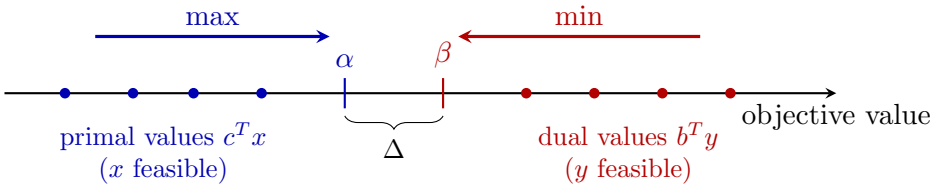


FIGURE 2. Weak duality. The maximization pushes the primal objective values $c^T x$ up from below, and the minimization pushes the dual objective values $b^T y$ down from above. Every primal value is at most every dual value, so $\alpha \le \beta$, with duality gap $\Delta = \beta - \alpha \ge 0$. For linear programs with a finite optimal value, the duality theorem (Theorem 8) gives $\Delta = 0$.

The remaining cases follow from these two facts. For example, if the primal is infeasible, then by fact i) the dual has no finite optimal value, so the dual is unbounded or infeasible. Both cases can occur. The following examples show the four combinations that are possible:

i) Both have optimal solutions: max x_1 s.t. $x_1 \le 1$. Dual: min y_1 s.t. $y_1 = 1$, $y_1 \ge 0$. Here $\alpha = \beta = 1$.

ii) Primal unbounded, dual infeasible: max x_1 s.t. $-x_1 \le 0$. Dual: min 0 s.t. $-y_1 = 1$, $y_1 \ge 0$.

iii) Primal infeasible, dual unbounded: max x_1 s.t. $x_1 \le -1$, $-x_1 \le -1$. Dual: min $(-y_1 - y_2)$ s.t. $y_1 - y_2 = 1$, $y \ge 0$. The points $y = (1+t, t)$ with $t \ge 0$ are feasible, and their objective value $-1 - 2t$ tends to $-\infty$.

iv) Both infeasible: max x_2 s.t. $x_1 \le -1$, $-x_1 \le -1$. The primal requires $x_1 \le -1$ and $x_1 \ge 1$. Dual: min $(-y_1 - y_2)$ s.t. $y_1 - y_2 = 0$, $0 = 1$, $y \ge 0$. The second dual constraint (for the variable x_2) cannot be satisfied.

Remark 9. *As we will see, the duality theorem does not hold for semidefinite programs: there, the optimal values of primal and dual can differ, and a finite optimal value need not be attained.*

1.5.3. Complementary slackness. We now characterize when a feasible pair (x, y) consists of optimal solutions.

THEOREM 10 (Complementary slackness, [12, Thm. 2.9]). *Let x be feasible for (P) and y be feasible for (D). Then x and y are optimal solutions if and only if the following equivalent conditions hold.*

i) Coordinatewise form:

$$y_i\,(b_i - a_i^T x) = 0 \quad \text{for all } i = 1, \dots, m\,. \tag{15}$$

ii) Matrix form:

$$y^T(b - Ax) = 0\,. \tag{16}$$

PROOF. For feasible x and y, the difference of the objective values is

$$b^T y - c^T x = y^T b - (A^T y)^T x = y^T(b - Ax) = \sum_{i=1}^{m} y_i\,(b_i - a_i^T x)\,. \tag{17}$$

Each term $y_i(b_i - a_i^T x)$ is non-negative, since $y_i \ge 0$ and $b_i - a_i^T x \ge 0$. So the sum is zero if and only if every term is zero, i.e. (16) and (15) are equivalent.

If x and y are optimal, then $c^T x = \alpha = \beta = b^T y$ by Theorem 8, so (17) is zero. Conversely, if (17) is zero, then $c^T x = b^T y$, and x and y are optimal by weak duality (Proposition 7). □

Complementary slackness is used to find or check optimal solutions: given an optimal x, the dual variables of the constraints that do not hold with equality are zero, which reduces the dual constraints $A^T y = c$ to a smaller system.

Example 11. Consider

$$\begin{aligned} &\max && x_1 + 2x_2 \\ &\text{subject to} && x_1 \le 1\,, \quad x_2 \le 1\,, \quad x_1 + x_2 \le \tfrac{3}{2}\,. \end{aligned} \tag{18}$$

The point $x = (\frac{1}{2}, 1)$ is feasible with objective value $\frac{5}{2}$. Is it optimal? We have $b - Ax = (\frac{1}{2}, 0, 0)$, so only the first constraint does not hold with equality. By complementary slackness, an optimal dual solution must have $y_1 = 0$. The dual constraints $A^T y = c$ read $y_1 + y_3 = 1$ and $y_2 + y_3 = 2$. With $y_1 = 0$ they give $y = (0, 1, 1) \ge 0$, so y is dual feasible. Its objective value is $b^T y = 0 + 1 + \frac{3}{2} = \frac{5}{2} = c^T x$. So x and y are optimal solutions, and $\alpha = \beta = \frac{5}{2}$. ◊

1.6. Infeasibility certificates

We now show how a vector $y \ge 0$ with $A^T y = \mathbf{0}$ and $b^T y < 0$ proves that the primal program is infeasible. For the feasibility problem ($c = 0$), such y is a dual feasible point with negative objective value. For general c, it need not be dual feasible. For this we will use Farkas lemma!

Let $y \in \mathbb{R}^m$ with $y \ge 0$, $A^T y = \mathbf{0}$, and $b^T y = -1$. Then the primal problem is infeasible. Such y is called an *infeasibility certificate*. In particular, it acts as an instance λ of Farkas Lemma (Theorem 3). Conversely, by Farkas Lemma (Theorem 3), every infeasible primal program has an infeasibility certificate.

i) For the feasibility problem ($c = 0$), such y is a feasible dual point with $b^T y = -1$. Then Cy is dual feasible for any $C > 0$, and $b^T(Cy) = -C \to -\infty$. So the dual is unbounded. By weak duality, any dual objective value is an upper bound on any primal objective value. This pushes the primal optimum to $-\infty$, and thus the primal problem is infeasible.

ii) For general c, if y_0 is dual feasible, then so is $y_0 + Cy$ for any $C > 0$, and $b^T(y_0 + Cy) = b^T y_0 - C \to -\infty$. The same argument applies.

iii) In case iii) of Section 1.5.2, $y = (1/2, 1/2)$ is an infeasibility certificate.

1.7. Summary

To summarize this chapter:

i) The feasible sets of linear programs are polyhedra, which are convex.

ii) Farkas Lemma: the system $Ax \le b$ has no solution if and only if some non-negative combination of its inequalities is the inequality $0 \le -1$.

iii) Every linear program can be brought into the standard form (P), which has the dual (D).

iv) Weak duality: dual objective values are upper bounds on primal objective values.

v) Duality theorem: if the primal or the dual has a finite optimal value, then both optimal values are attained and equal. If the primal or the dual is unbounded, then the other is infeasible.

vi) Complementary slackness: feasible x and y are optimal if and only if $y^T(b - Ax) = 0$, i.e. $y_i(b_i - a_i^T x) = 0$ for all i.

vii) An infeasible primal has an infeasibility certificate $y \ge 0$ with $A^T y = \mathbf{0}$ and $b^T y = -1$.

CHAPTER 2

Linear programs in combinatorial optimization

The aim of this chapter is to use linear programs to bound graph parameters. For each problem we write it as an integer linear program, relax it to a linear program, and compare the two optimal values. This approach is described in [12, Sec. 1 and 2.3]. Sections marked with $*$ were not treated in class. The chapter contains:

i) integer linear programs and LP relaxations (Section 2.2);

ii) stable sets and the polytopes $\mathrm{STAB}(G) \subseteq \mathrm{QSTAB}(G) \subseteq \mathrm{FSTAB}(G)$ (Section 2.3);

iii) matchings and vertex covers, whose LP relaxations are dual to each other, König's theorem (stated without proof), and LP rounding for vertex cover, a first example of an approximation algorithm (Section 2.4);

iv) flows from a source to a target, and the maximum flow problem as a linear program (Section 2.5);

v) Chvátal–Gomory cuts, the Chvátal closure, and the Chvátal rank (Section 2.6).

2.1. Graphs

A graph $G = (V, E)$ consists of a finite set V of vertices and a set E of edges $e = \{v, w\}$ with $v, w \in V$, $v \neq w$. A graph is *bipartite* if $V = V_1 \cup V_2$ with $V_1 \cap V_2 = \emptyset$ such that every edge has one endpoint in V_1 and one in V_2. A *clique* is a set of pairwise adjacent vertices. K_n is the complete graph on n vertices; K_3 is the triangle.

We index vectors by vertices or edges: $x \in \mathbb{R}^V$ or $x \in \mathbb{R}^E$. For $S \subseteq V$, the incidence vector $\chi^S \in \{0,1\}^V$ has $\chi^S_v = 1$ if $v \in S$ and 0 otherwise; similarly $\chi^F \in \{0,1\}^E$ for $F \subseteq E$. For $x \in \mathbb{R}^V$ and $S \subseteq V$ we write $x(S) = \sum_{v \in S} x_v$, and similarly for edges.

2.2. Integer linear programs and LP relaxations

An integer linear program is

$$\text{(IP)} \qquad \max\{c^T x : Ax \leq b,\ x \in \mathbb{Z}^n\}\,.$$

We now relax this to the *LP relaxation*

$$\text{(LP)} \qquad \max\{c^T x : Ax \leq b,\ x \in \mathbb{R}^n\}\,,$$

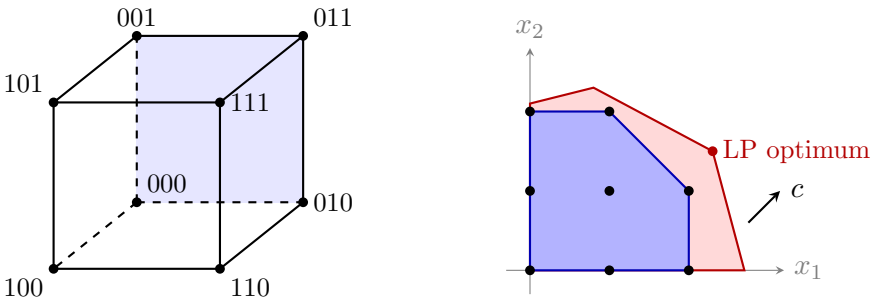
the linear program (P) obtained by dropping the constraint $x \in \mathbb{Z}^n$. Solving integer linear programs is NP-hard in general [9], while linear programs can be solved efficiently [12, Sec. 2.2].

The idea of a relaxation. A relaxation drops a constraint, here $x \in \mathbb{Z}^n$. This makes the feasible set larger: every feasible point of (IP) is a feasible point of (LP). In general, a program 2 with the same objective function is a *relaxation* of a program 1 if every feasible point of program 1 is feasible for program 2. Optimizing over a larger set gives a result that is at least as good. Writing $\mathrm{Opt}(\cdot)$ for the optimal value of a program,

$$\begin{aligned} &\mathrm{Opt(IP)} \leq \mathrm{Opt(LP)} && \text{for maximization problems,} \\ &\mathrm{Opt(LP)} \leq \mathrm{Opt(IP)} && \text{for minimization problems.} \end{aligned} \tag{19}$$

So solving the LP relaxation, which is easy, gives an upper bound (for maximization) or a lower bound (for minimization) on the optimal value of the integer program, which is hard to compute. See Fig. 1(b). All results in this chapter are built on this idea.

For some problems the LP relaxation has an integral optimal solution, and then the LP relaxation and the integer program have the same optimal value. Below we see examples: stable sets and matchings in bipartite graphs, and flows.



(a) $[0,1]^3$ (b) relaxation

FIGURE 1. (a) The hypercube $[0,1]^3$. Its 8 vertices are the points of $\{0,1\}^3$, labelled by their entries. In general, $[0,1]^n$ has 2^n vertices. (b) The idea of a relaxation, for $\max\{x_1+x_2 : Ax \le b,\ x \in \mathbb{Z}^2\}$ with $c = (1,1)$. Black dots: feasible points of (IP). Blue: their convex hull. Red: the larger feasible set of (LP). Here $\mathrm{Opt(IP)} = 3$ (at $(2,1)$ and $(1,2)$) and $\mathrm{Opt(LP)} = 3.8$ (at the red vertex $(2.3, 1.5)$), so $\mathrm{Opt(IP)} \le \mathrm{Opt(LP)}$.

Binary optimization. An important special case are *binary* programs, where $x \in \{0,1\}^n$. The 2^n points of $\{0,1\}^n$ are the vertices of the hypercube $[0,1]^n = \{x \in \mathbb{R}^n : 0 \le x \le \mathbf{1}\}$, see Fig. 1(a). So a binary program optimizes a linear function over those vertices of the hypercube that satisfy $Ax \le b$. Its optimal value equals the optimal value of the linear program over the convex hull of these vertices, since a linear function attains its maximum over a convex hull at one of the generating points. In practice, neither approach works for large n:

i) Checking all vertices takes 2^n steps; for $n = 100$ this is about 10^{30}.

ii) The convex hull of the feasible vertices is a polytope (Section 1.2), but in general it has exponentially many faces, so its description by linear inequalities is exponentially large and usually not known explicitly [**12**, Sec. 2.3].

The LP relaxation replaces this convex hull by the hypercube cut by $Ax \le b$, which has only $m + 2n$ inequalities.

2.3. Stable sets

A set $S \subseteq V$ is *stable* (or independent) if no edge has both endpoints in S. The stability number $\alpha(G)$ is the largest size of a stable set. See Fig. 2.

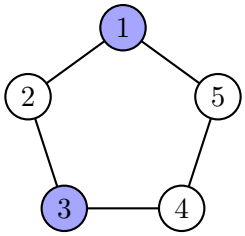


FIGURE 2. The 5-cycle C_5. The filled vertices form the stable set $S = \{1,3\}$: no edge joins two filled vertices. Every set of three vertices of C_5 contains an edge, so $\alpha(C_5) = 2$.

Computing $\alpha(G)$ is NP-hard [**9**], see also [**12**, Sec. 2.3]. As an integer program,

$$\text{(20)} \qquad \text{(IP)} \qquad \alpha(G) = \max\{\sum_{v \in V} x_v : x_v + x_w \le 1 \text{ for all } \{v,w\} \in E,\ 0 \le x \le \mathbf{1},\ x \in \mathbb{Z}^V\}\,.$$

The feasible points of (IP) are exactly the incidence vectors χ^S of stable sets S. So (IP) is a binary program: it optimizes over vertices of the hypercube $[0,1]^V$ (Section 2.2).

The stable set polytope. The convex hull of the feasible points of (IP) is the *stable set polytope*

$$\mathrm{STAB}(G) = \mathrm{conv}\{\chi^S : S \subseteq V \text{ stable}\}\,, \tag{21}$$

and $\alpha(G) = \max\{\sum_{v\in V} x_v : x \in \mathrm{STAB}(G)\}$.

STAB(G) is a polytope. So there exists a system of linear inequalities whose solution set is STAB(G) (Section 1.2). But this system is in general exponentially large and not known explicitly [**12**, Sec. 2.3].

LP relaxations. Every polytope that contains STAB(G) gives an LP relaxation of (IP). So there is not one LP relaxation, but many. We consider two, described by known inequalities that are valid for STAB(G):

$$\begin{aligned} \mathrm{FSTAB}(G) &= \{x \in \mathbb{R}^V : 0 \le x \le \mathbf{1},\ x_v + x_w \le 1 \text{ for all } \{v,w\} \in E\}\,, \\ \mathrm{QSTAB}(G) &= \{x \in \mathbb{R}^V : x \ge 0,\ x(K) \le 1 \text{ for all cliques } K\}\,. \end{aligned} \tag{22}$$

FSTAB(G) is obtained from (IP) by dropping the constraint $x \in \mathbb{Z}^V$. QSTAB(G) uses the clique inequalities instead of the edge inequalities. We now relax (IP) to the two linear programs

$$\begin{aligned} (\mathrm{LP_F}) \qquad & \max\{\sum_{v\in V} x_v : x \in \mathrm{FSTAB}(G)\}\,, \\ (\mathrm{LP_Q}) \qquad & \max\{\sum_{v\in V} x_v : x \in \mathrm{QSTAB}(G)\}\,. \end{aligned} \tag{23}$$

These polytopes are defined as in [**12**, Sec. 2.3]. There, FSTAB(G) has no upper bounds $x \le \mathbf{1}$, since graphs are assumed to have no isolated vertices.

We have

$$\mathrm{STAB}(G) \subseteq \mathrm{QSTAB}(G) \subseteq \mathrm{FSTAB}(G)\,. \tag{24}$$

That is, every point of STAB(G) is feasible for QSTAB(G), and every point of QSTAB(G) is feasible for FSTAB(G). So (LP$_\mathrm{Q}$) is a relaxation of (IP), and (LP$_\mathrm{F}$) is a relaxation of (LP$_\mathrm{Q}$). By (19),

$$\alpha(G) = \mathrm{Opt(IP)} \le \mathrm{Opt(LP_Q)} \le \mathrm{Opt(LP_F)}\,. \tag{25}$$

Further constraints. More linear inequalities that hold for every $x \in \mathrm{STAB}(G)$ give smaller polytopes containing STAB(G), and hence smaller upper bounds on $\alpha(G)$ [**12**, Sec. 2.3]. A *hole* is a set $C \subseteq V$ of at least 5 vertices whose induced subgraph is a cycle (a cycle without chords). An *antihole* is a set $B \subseteq V$ of at least 5 vertices whose induced subgraph is the complement of a cycle.

i) Odd hole constraints: for every hole C with $|C|$ odd,

$$x(C) \le \tfrac{|C|-1}{2}\,. \tag{26}$$

They are valid since a stable set contains at most $\frac{|C|-1}{2}$ vertices of an odd cycle of length $|C|$.

ii) Odd antihole constraints: for every antihole B with $|B|$ odd,

$$x(B) \le 2\,. \tag{27}$$

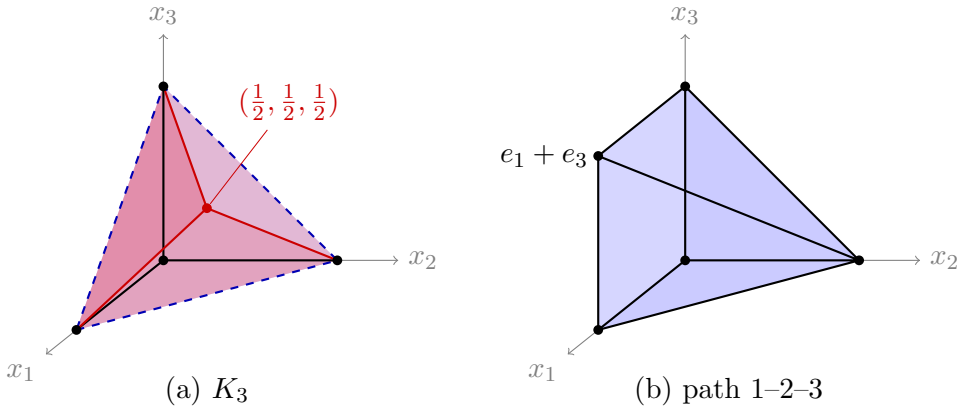
They are valid since a stable set in the complement of a cycle is a set of pairwise adjacent vertices of the cycle, and a cycle of length at least 5 has no three pairwise adjacent vertices.

For example, for the 5-cycle C_5 (Fig. 2), the odd hole constraint is $x_1 + \cdots + x_5 \le 2$. It is violated by $\frac{1}{2}\mathbf{1} \in \mathrm{FSTAB}(C_5)$, and it gives the upper bound $2 = \alpha(C_5)$.

Known results:

i) STAB(G) = FSTAB(G) if and only if G is bipartite [**12**, Prop. 2.10], [**6**, Ch. 9].

ii) Optimizing a linear function over QSTAB(G) is NP-hard in general [6, Ch. 9]. The Lovász ϑ function gives a convex set between STAB(G) and QSTAB(G) over which one can optimize efficiently [6, Ch. 9]. We will see this in the chapter on semidefinite programming.



(a) K_3 (b) path 1–2–3

FIGURE 3. Black dots: incidence vectors of stable sets. (a) For the triangle K_3, $\mathrm{STAB}(K_3) = \mathrm{QSTAB}(K_3)$ is the simplex $\mathrm{conv}\{\mathbf{0}, e_1, e_2, e_3\}$ (blue). $\mathrm{FSTAB}(K_3)$ additionally contains the pyramid over the face $\mathrm{conv}\{e_1, e_2, e_3\}$ (dashed) with apex $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$ (red), cf. [12, Fig. 5]. (b) For the path with edges $\{1,2\}$, $\{2,3\}$, $\mathrm{STAB} = \mathrm{QSTAB} = \mathrm{FSTAB} = \mathrm{conv}\{\mathbf{0}, e_1, e_2, e_3, e_1+e_3\}$.

Example 12. See Fig. 3 for the triangle and the path.

i) Triangle K_3 with edges $\{1,2\}$, $\{2,3\}$, $\{1,3\}$. The stable sets are $\emptyset$, $\{1\}$, $\{2\}$, $\{3\}$, so $\alpha(K_3) = 1$. The three polytopes are

$$\begin{aligned} \mathrm{STAB}(K_3) &= \mathrm{conv}\{\mathbf{0}, e_1, e_2, e_3\}\,, \\ \mathrm{FSTAB}(K_3) &= \{x : 0 \le x \le \mathbf{1},\ x_1 + x_2 \le 1,\ x_2 + x_3 \le 1,\ x_1 + x_3 \le 1\}\,, \\ \mathrm{QSTAB}(K_3) &= \{x \ge 0 : x_1 + x_2 + x_3 \le 1\}\,. \end{aligned} \tag{28}$$

ii) The path with vertices $1,2,3$ and edges $\{1,2\}$, $\{2,3\}$. The stable sets are $\emptyset$, $\{1\}$, $\{2\}$, $\{3\}$, $\{1,3\}$, so $\alpha = 2$. The graph is bipartite, and

$$\mathrm{STAB} = \mathrm{QSTAB} = \mathrm{FSTAB} = \{x : 0 \le x \le \mathbf{1},\ x_1 + x_2 \le 1,\ x_2 + x_3 \le 1\}\,. \tag{29}$$

iii) Complete graph K_n, $n \ge 2$. Here $\alpha(K_n) = 1$, and $\max_{x \in \mathrm{QSTAB}(K_n)} \sum_{v \in V} x_v = 1$. But $\frac{1}{2}\mathbf{1} \in \mathrm{FSTAB}(K_n)$, so $\max_{x \in \mathrm{FSTAB}(K_n)} \sum_{v \in V} x_v \ge n/2$. So the upper bound on $\alpha(G)$ from $\mathrm{FSTAB}(G)$ can exceed $\alpha(G)$ by a factor of $n/2$.

◊

To summarize this section: the LPs over $\mathrm{FSTAB}(G)$ and $\mathrm{QSTAB}(G)$ give upper bounds on the stability number $\alpha(G)$. For bipartite graphs the upper bound from $\mathrm{FSTAB}(G)$ equals $\alpha(G)$; for K_n it is $n/2$, while $\alpha(K_n) = 1$.

2.4. Matchings and vertex covers

The aim of this section is to compare the matching number and the vertex cover number of a graph, using that their LP relaxations are dual to each other.

2.4.1. Matchings. A *matching* is a set $M \subseteq E$ of edges such that no two edges in M share a vertex. The matching number $\nu(G)$ is the largest size of a matching.

Example 13 (Workers and tasks)**.** Matchings in bipartite graphs model assignments. Let V_1 be a set of n workers and V_2 a set of n tasks, and let $\{v, w\} \in E$ with $v \in V_1$, $w \in V_2$ if worker v is qualified for task w. This is a bipartite graph. A matching assigns each worker at most one task, and each task to at most one worker. ◊

We use one variable x_{uv} for each edge $\{u,v\} \in E$ (so $x_{uv} = x_{vu}$), with $x_{uv} = 1$ if the edge is in the matching and $x_{uv} = 0$ otherwise. The *fractional matching polytope* is

$$P_{\mathrm{M}}(G) = \Big\{ x \in \mathbb{R}^E : \sum_{u\,:\,\{u,v\}\in E} x_{uv} \le 1 \ \text{ for all } v \in V, \quad x_{uv} \ge 0 \ \text{ for all } \{u,v\} \in E \Big\}. \tag{30}$$

The constraint for v says that at most one edge at v is chosen. Given weights w_{uv} on the edges, a *maximum weight matching* is an optimal solution of

$$\text{(IP)} \qquad \max\Big\{ \sum_{\{u,v\}\in E} w_{uv} x_{uv} : x \in P_{\mathrm{M}}(G),\ x \in \mathbb{Z}^E \Big\}.$$

The feasible points of (IP) are the incidence vectors of matchings. We now relax this to

$$\text{(LP)} \qquad \max\Big\{ \sum_{\{u,v\}\in E} w_{uv} x_{uv} : x \in P_{\mathrm{M}}(G) \Big\}.$$

For $w_{uv} = 1$ on all edges, $\mathrm{Opt(IP)} = \nu(G)$; we write $\nu^*(G)$ for $\mathrm{Opt(LP)}$ in this case.

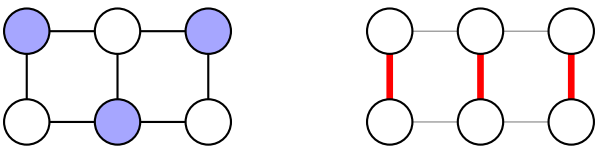
For general graphs, the vertices of $P_{\mathrm{M}}(G)$ need not be integral, but they are half-integral.

THEOREM 14 (Balinski [1]; see also [13]). *Let G be any graph. Every vertex of the fractional matching polytope $P_{\mathrm{M}}(G)$ in (30) has all entries in $\{0, \frac{1}{2}, 1\}$.*

We do not prove this here. For the triangle K_3, the vertex $x = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is half-integral but not integral.

A matching is *perfect* if it covers every vertex. A maximum weight perfect matching is obtained by replacing the constraints $\sum_{u:\{u,v\}\in E} x_{uv} \le 1$ in $P_{\mathrm{M}}(G)$ by equalities $\sum_{u:\{u,v\}\in E} x_{uv} = 1$ for all $v \in V$. For bipartite graphs, (LP) has an integral optimal solution whenever it has an optimal solution [16, Ch. 18].

2.4.2. Vertex covers. A *vertex cover* is a set $C \subseteq V$ such that every edge has at least one endpoint in C. The vertex cover number $\tau(G)$ is the smallest size of a vertex cover. See Fig. 4 for a vertex cover and a matching.



(a) vertex cover (b) matching

FIGURE 4. A bipartite graph with 6 vertices and 7 edges. (a) The filled vertices form a vertex cover: every edge has at least one filled endpoint, i.e. every edge is "seen" by a filled vertex. (b) The thick edges form a matching: they pair up vertices, and no two of them share a vertex. Here $\nu(G) = \tau(G) = 3$.

We use one variable y_v for each vertex $v \in V$, with $y_v = 1$ if v is in the vertex cover. The *fractional vertex cover polyhedron* is

$$P_{\mathrm{VC}}(G) = \big\{ y \in \mathbb{R}^V : y_u + y_v \ge 1 \ \text{ for all } \{u,v\} \in E, \quad y_v \ge 0 \ \text{ for all } v \in V \big\}. \tag{31}$$

The constraint for $\{u,v\}$ says that the edge has at least one endpoint in the vertex cover. The integer program is

$$\text{(IP)} \qquad \tau(G) = \min\Big\{ \sum_{v\in V} y_v : y \in P_{\mathrm{VC}}(G),\ y \in \mathbb{Z}^V \Big\}.$$

An optimal solution of (IP) has entries in $\{0,1\}$ (replacing entries larger than 1 by 1 keeps feasibility), and is the incidence vector of a vertex cover. We now relax this to

$$\text{(LP)} \qquad \tau^*(G) = \min\Big\{ \sum_{v\in V} y_v : y \in P_{\mathrm{VC}}(G) \Big\}.$$

As for matchings, the vertices of $P_{\mathrm{VC}}(G)$ are half-integral.

THEOREM 15 (Nemhauser–Trotter [14]). *Let G be any graph. Every vertex of the fractional vertex cover polyhedron $P_{\mathrm{VC}}(G)$ in (31) has all entries in $\{0, \frac{1}{2}, 1\}$. In particular, (LP) for $\tau^*(G)$ has an optimal solution with entries in $\{0, \frac{1}{2}, 1\}$.*

We do not prove this here. For example, for the triangle K_3 the optimal solution $y = \frac{1}{2}\mathbf{1}$ of (LP) is half-integral (Section 2.4.4).

2.4.3. LP rounding for vertex cover. Computing $\tau(G)$ is NP-hard [9], but (LP) can be solved efficiently [12, Sec. 2.2]. We now turn an optimal solution of (LP) into a vertex cover that is at most twice as large as a smallest vertex cover [8].

Definition 16. *An algorithm for a minimization problem has* approximation ratio $\rho \geq 1$ *if, for every input, it returns a feasible solution whose objective value is at most ρ times the optimal value.*

Rounding. Let y^* be an optimal solution of (LP), so $\sum_{v \in V} y_v^* = \tau^*(G)$. Round up the entries that are at least $\frac{1}{2}$:

$$C = \{v \in V : y_v^* \geq \tfrac{1}{2}\}\,. \tag{32}$$

Proposition 17. *C is a vertex cover, and $|C| \leq 2\,\tau^*(G) \leq 2\,\tau(G)$. So LP rounding has approximation ratio 2.*

PROOF. Let $\{v, w\} \in E$. Since y^* is feasible, $y_v^* + y_w^* \geq 1$, so $y_v^* \geq \frac{1}{2}$ or $y_w^* \geq \frac{1}{2}$. Hence $v \in C$ or $w \in C$, and C is a vertex cover. For $v \in C$ we have $1 \leq 2y_v^*$. Using $y^* \geq 0$,

$$|C| = \sum_{v \in C} 1 \leq \sum_{v \in C} 2y_v^* \leq 2 \sum_{v \in V} y_v^* = 2\,\tau^*(G) \leq 2\,\tau(G)\,, \tag{33}$$

where the last inequality is $\mathrm{Opt(LP)} \leq \mathrm{Opt(IP)}$ from (19), since vertex cover is a minimization problem. □

By Theorem 15 we may take y^* with entries in $\{0, \frac{1}{2}, 1\}$; then rounding means: take all vertices with $y_v^* = \frac{1}{2}$ or $y_v^* = 1$.

Example 18. For the triangle K_3: $\mathrm{Opt(LP)} = \tau^*(K_3) = \frac{3}{2}$ with $y^* = \frac{1}{2}\mathbf{1}$, and $\mathrm{Opt(IP)} = \tau(K_3) = 2$. Rounding gives $C = \{1, 2, 3\}$ with $|C| = 3 \leq 2\,\tau(K_3) = 4$. ◊

2.4.4. Duality of the LP relaxations*. *(Not treated in class.)*

We show that the LP relaxations with optimal values $\nu^*(G)$ and $\tau^*(G)$ are dual to each other, using the recipe of Section 1.4.3. Take the matching relaxation with $w_{uv} = 1$. In standard form (P) it has the objective $\sum_{\{u,v\} \in E} x_{uv}$ and two kinds of inequalities:

$$\sum_{u\,:\,\{u,v\} \in E} x_{uv} \leq 1 \quad \text{for each } v \in V\,, \qquad -x_{uv} \leq 0 \quad \text{for each } \{u, v\} \in E\,. \tag{34}$$

The dual (D) has one variable for each primal inequality: $y_v \geq 0$ for each vertex v, and $s_{uv} \geq 0$ for each edge $\{u, v\}$. Its objective $b^T y$ has the right-hand sides as coefficients: $\sum_{v \in V} y_v \cdot 1 + \sum_{\{u,v\} \in E} s_{uv} \cdot 0 = \sum_{v \in V} y_v$. It has one equality constraint for each primal variable x_{uv} (a row of $A^T y = c$). The variable x_{uv} appears with coefficient 1 in the inequalities of u and of v, with coefficient -1 in $-x_{uv} \leq 0$, and with coefficient 1 in the objective. So the dual is

$$\min \sum_{v \in V} y_v \quad \text{s.t.} \quad y_u + y_v - s_{uv} = 1 \ \text{ for all } \{u, v\} \in E\,, \quad y \geq 0\,, \quad s \geq 0\,. \tag{35}$$

Since $s_{uv} = y_u + y_v - 1$, the constraint $s_{uv} \geq 0$ says $y_u + y_v \geq 1$. So the dual is the vertex cover relaxation. Both programs are feasible ($x = \mathbf{0}$, $y = \mathbf{1}$), so by weak duality and Theorem 8,

$$\nu(G) \leq \nu^*(G) = \tau^*(G) \leq \tau(G)\,. \tag{36}$$

Example 19. For the triangle K_3: $\nu(K_3) = 1$ and $\tau(K_3) = 2$. The points $x = \frac{1}{2}\mathbf{1} \in \mathbb{R}^E$ and $y = \frac{1}{2}\mathbf{1} \in \mathbb{R}^V$ are feasible for the two relaxations with the same objective value $\frac{3}{2}$. By weak duality, both are optimal, and $\nu^*(K_3) = \tau^*(K_3) = \frac{3}{2}$. ◊

2.4.5. Bipartite graphs. In every graph, $\nu(G) \le \tau(G)$: a vertex cover must contain an endpoint of each edge of a matching, and these endpoints are distinct since the edges of a matching are disjoint. For bipartite graphs, such as the workers and tasks of Section 2.4, the matching number equals the vertex cover number, and both integer programs have the same optimal value as their LP relaxations.

THEOREM 20 (König [10]). *If G is bipartite, then $\nu(G) = \tau(G)$: the largest size of a matching equals the smallest size of a vertex cover.*

THEOREM 21 (see [16, Ch. 18]). *If G is bipartite, then the integer programs for matching and vertex cover have the same optimal values as their LP relaxations: $\nu(G) = \nu^*(G)$ and $\tau(G) = \tau^*(G)$.*

We do not prove these theorems here; see [16, Ch. 16 and 18]. Theorem 21 and LP duality (Section 2.4.4) give Theorem 20: $\nu(G) = \nu^*(G) = \tau^*(G) = \tau(G)$.

2.4.6. Edmonds' characterization of the matching polytope*. *(Not treated in class.)*

For general graphs, the matching relaxation is not integral, as the triangle shows. Edmonds [3] showed that the matching polytope $\operatorname{conv}\{\chi^M : M \text{ matching}\}$ equals the set of $x \in P_{\mathrm{M}}(G)$ (see (30)) that satisfy the *odd set inequalities*

$$\sum_{\{u,v\}\in E\,:\,u,v\in U} x_{uv} \le \tfrac{|U|-1}{2} \qquad \text{for all } U \subseteq V \text{ with } |U| \text{ odd}\,. \tag{37}$$

The sum runs over the edges with both endpoints in U. We do not prove this here. These odd set inequalities are examples of Chvátal–Gomory cuts (Section 2.6).

To summarize this section: the matching number and the vertex cover number satisfy $\nu(G) \le \nu^*(G) = \tau^*(G) \le \tau(G)$, where the middle equality is LP duality. For bipartite graphs all four numbers are equal. Rounding an optimal LP solution gives a vertex cover of size at most $2\tau(G)$.

2.5. Flows

The aim of this section is to write the maximum flow problem as a linear program.

A directed graph $D = (V, E)$ has a set of directed edges $E \subseteq V \times V$; the directed edge (u, v) goes from u to v. For a vector $f \in \mathbb{R}^E$ we write f_{uv} for its entry at $(u, v) \in E$. For a vertex v, the directed edges entering v are the $(u, v) \in E$, and the directed edges leaving v are the $(v, w) \in E$. Fix two vertices, a *source* s and a *target* (or *sink*) $t \ne s$, and capacities $c \in \mathbb{R}^E_+$ on the directed edges.

An *s-t flow* is a vector $f \in \mathbb{R}^E$ with

$$\begin{aligned} 0 \le f_{uv} \le c_{uv} \quad &\text{for all } (u,v) \in E\,, \\ \sum_{u\,:\,(u,v)\in E} f_{uv} = \sum_{w\,:\,(v,w)\in E} f_{vw} \quad &\text{for all } v \in V \setminus \{s,t\}\,. \end{aligned} \tag{38}$$

The first condition says that the flow on each directed edge is non-negative and at most its capacity. The second condition (flow conservation) says that at every vertex other than source and target, the flow entering equals the flow leaving. The value of the flow is the net flow leaving the source,

$$\operatorname{val}(f) = \sum_{w\,:\,(s,w)\in E} f_{sw} - \sum_{u\,:\,(u,s)\in E} f_{us}\,. \tag{39}$$

See Fig. 5.

The maximum flow problem is the linear program

$$\max \ \operatorname{val}(f) \quad \text{s.t.} \quad f \text{ is an } s\text{-}t \text{ flow}\,. \tag{40}$$

It is feasible ($f = \mathbf{0}$) and bounded ($\operatorname{val}(f) \le \sum_{w:(s,w)\in E} c_{sw}$), so by Theorem 8 it has an optimal solution. If all capacities are integers, then there is a maximum flow with integer values [5].

Exercise 22. *Bring the maximum flow problem into the standard form (P) and compute its dual.*

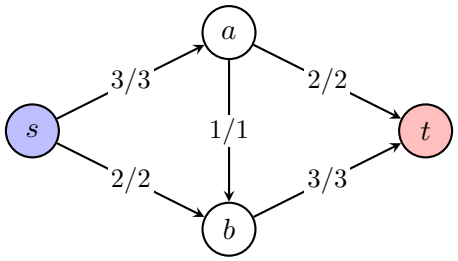


FIGURE 5. A flow from the source s to the target t. Each directed edge (u, v) is labelled f_{uv}/c_{uv} (flow/capacity). Flow conservation holds at a (in: 3, out: $1+2$) and at b (in: $2+1$, out: 3). The value of the flow is $3+2=5$. No flow has larger value, since the two directed edges leaving s have total capacity 5.

To summarize this section: a flow from the source to the target respects the capacities and flow conservation, and maximizing its value is a linear program.

2.6. Chvátal–Gomory cuts

The aim of this section is to strengthen an LP relaxation by adding inequalities that are valid for all integer feasible points, but cut off fractional points. We follow [11, Sec. 3.2].

Let $Q = \{x \in \mathbb{R}^n : Ax \le b\}$ be the feasible set of the LP relaxation of an integer program, and let $Q_I = \operatorname{conv}(Q \cap \mathbb{Z}^n)$ be the convex hull of its integer points. Maximizing a linear function over Q_I gives the optimal value of the integer program, since a linear function attains its maximum over a convex hull at one of the generating points. In general $Q_I \subsetneq Q$, and we want to find linear inequalities that cut Q down towards Q_I.

2.6.1. Chvátal–Gomory cuts. By (2), for every $\lambda \in \mathbb{R}^m$ with $\lambda \ge 0$ the inequality

$$(\lambda^T A)\, x \le \lambda^T b \tag{41}$$

is valid for all $x \in Q$. Suppose that the coefficient vector $\lambda^T A$ is integral. Then for every integer point $x \in Q \cap \mathbb{Z}^n$ the left-hand side $(\lambda^T A)\, x$ is an integer, and an integer that is at most $\lambda^T b$ is at most $\lfloor \lambda^T b \rfloor$. So the *Chvátal–Gomory cut*

$$(\lambda^T A)\, x \le \lfloor \lambda^T b \rfloor \tag{42}$$

is valid for all points of $Q \cap \mathbb{Z}^n$, and hence for Q_I. It may cut off fractional points of Q. This rounding idea is due to Gomory [7] and Chvátal [2].

Example 23. Let $Q = \{x \in \mathbb{R}^2 : x_1 \ge 0,\ x_2 \ge 0,\ 2x_1 + 2x_2 \le 3\}$. Its integer points are $(0,0)$, $(1,0)$, $(0,1)$. With $\lambda = \frac{1}{2}$ on the inequality $2x_1 + 2x_2 \le 3$ we get $x_1 + x_2 \le \frac{3}{2}$, with integral coefficients. Rounding down gives the cut $x_1 + x_2 \le 1$. It cuts off the fractional vertices $(\frac{3}{2}, 0)$ and $(0, \frac{3}{2})$ of Q, and together with $x \ge 0$ it describes $Q_I = \operatorname{conv}\{(0,0),(1,0),(0,1)\}$. See Fig. 6. ◊

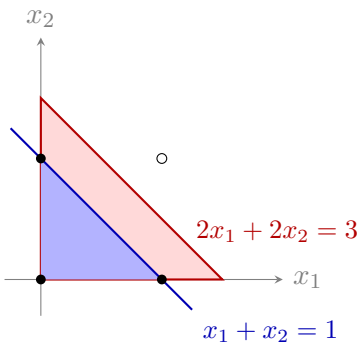


FIGURE 6. A Chvátal–Gomory cut. Red: $Q = \{x \ge 0 : 2x_1 + 2x_2 \le 3\}$. Dividing $2x_1 + 2x_2 \le 3$ by 2 and rounding down the right-hand side gives the cut $x_1 + x_2 \le 1$ (blue line), which is valid for the integer points (black dots) of Q and cuts off the fractional vertices $(\frac{3}{2}, 0)$ and $(0, \frac{3}{2})$. The integer point $(1,1)$ (open circle) is not in Q. Blue: Q_I.

Several inequalities from the previous sections are Chvátal–Gomory cuts of the LP relaxations.

i) Clique inequality of the triangle. Sum the three edge inequalities of K_3 with $\lambda = \frac{1}{2}$ each: $x_1 + x_2 + x_3 \le \frac{3}{2}$. Rounding down gives $x_1 + x_2 + x_3 \le 1$, the clique inequality. Together with $\mathrm{FSTAB}(K_3)$ it describes $\mathrm{STAB}(K_3)$ (Section 2.3).

ii) Odd hole constraints. For a hole C with $|C|$ odd, sum the $|C|$ edge inequalities of the cycle with $\lambda = \frac{1}{2}$ each. Every vertex of C lies on two edges of the cycle, so this gives $x(C) \le \frac{|C|}{2}$. Rounding down gives $x(C) \le \frac{|C|-1}{2}$, the odd hole constraint.

iii) Edmonds' odd set inequalities for matchings (Section 2.4). Let $U \subseteq V$ with $|U|$ odd. Sum the constraints $\sum_{u:\{u,v\}\in E} x_{uv} \le 1$ over $v \in U$. An edge with both endpoints in U appears twice, and an edge with exactly one endpoint in U appears once, so

$$2 \sum_{\{u,v\}\in E\,:\,u,v\in U} x_{uv} \;+ \sum_{\{u,v\}\in E\,:\,u\in U,\ v\notin U} x_{uv} \;\le\; |U|\,. \tag{43}$$

Adding the inequalities $-x_{uv} \le 0$ for the edges with exactly one endpoint in U, multiplying by $\frac{1}{2}$, and rounding down the right-hand side gives

$$\sum_{\{u,v\}\in E\,:\,u,v\in U} x_{uv} \le \left\lfloor \frac{|U|}{2} \right\rfloor = \frac{|U|-1}{2}\,. \tag{44}$$

2.6.2. Chvátal closure and Chvátal rank. The *Chvátal closure* of Q is the set of points that satisfy all Chvátal–Gomory cuts,

$$Q' = \{x \in \mathbb{R}^n : (\lambda^T A)\, x \le \lfloor \lambda^T b \rfloor \text{ for all } \lambda \ge 0 \text{ with } \lambda^T A \text{ integral}\}\,. \tag{45}$$

Taking $\lambda = e_i$ shows that Q' satisfies the inequalities of Q whenever the rows of A are integral, and every Chvátal–Gomory cut is valid for Q_I. So

$$Q_I \subseteq Q' \subseteq Q\,. \tag{46}$$

One can repeat this: $Q^{(1)} = Q'$ and $Q^{(t+1)} = (Q^{(t)})'$.

THEOREM 24 (Chvátal [2], see [11, Sec. 3.2]). *Let $Q = \{x : Ax \le b\}$ be a polytope with A and b rational. Then Q' is a polytope, and $Q^{(t)} = Q_I$ for some finite t.*

We do not prove this here. The smallest such t is the *Chvátal rank* of Q. In the examples above the rank is 1: one round of cuts gives Q_I. For example, FSTAB(K_3) and the polytope $\{x \in [0,1]^n : x_1 + \cdots + x_n \ge \frac{1}{2}\}$ have Chvátal rank 1; for the latter, the cut $x_1 + \cdots + x_n \ge 1$ gives the convex hull of its integer points [11, Sec. 3.4]. In general, the Chvátal rank can be large. For polytopes contained in $[0,1]^n$ it is at most of order $n^2 \log n$, and optimizing a linear function over the first Chvátal closure Q' is already hard in general [11, Sec. 3.2].

To summarize this section: rounding down the right-hand side of a valid inequality with integral coefficients gives a Chvátal–Gomory cut, which is valid for all integer points and can cut off fractional points. The clique inequality of the triangle, the odd hole constraints, and Edmonds' odd set inequalities all arise in this way.

2.7. Summary

To summarize this chapter:

i) A relaxation enlarges the feasible set. So Opt(IP) $\le$ Opt(LP) for maximization problems and Opt(LP) $\le$ Opt(IP) for minimization problems.

ii) Stable sets: STAB(G) $\subseteq$ QSTAB(G) $\subseteq$ FSTAB(G), so LPs over FSTAB(G) and QSTAB(G) give upper bounds on $\alpha(G)$. STAB(G) = FSTAB(G) if and only if G is bipartite.

iii) Matchings and vertex covers: the two LP relaxations are dual to each other, so $\nu(G) \le \nu^*(G) = \tau^*(G) \le \tau(G)$. For bipartite graphs all four numbers are equal (König's theorem). For general graphs, Edmonds' odd set inequalities describe the matching polytope. Rounding an optimal solution of the vertex cover LP gives a 2-approximation for $\tau(G)$.

iv) Flows: the maximum flow problem is a linear program. For integer capacities it has an optimal solution with integer values.

v) Chvátal–Gomory cuts: if $a^T x \le \gamma$ is valid for the LP relaxation and a is integral, then $a^T x \le \lfloor \gamma \rfloor$ is valid for all integer feasible points.

CHAPTER 3

Review linear algebra

The aim of this chapter is to review the linear algebra used in the other chapters, in particular for positive semidefinite matrices. We generally consider real vector spaces, except for a few examples on Pauli matrices and qubits. The chapter contains:

i) vector spaces, bases, and dimension (Section 3.1);
ii) vectors: inner product, outer product, and expansion in an orthonormal basis (Section 3.2);
iii) matrices: products, trace, trace inner product, and projectors (Section 3.3);
iv) eigenvalues, eigenvectors, and the spectral theorem (Section 3.4);
v) positive semidefinite matrices and equivalent definitions (Section 3.5);
vi) the Löwner order, the PSD cone, and dual cones (Section 3.6).

3.1. Vector spaces

3.1.1. Definition. A real vector space is a set V with an addition $+\colon V \times V \to V$ and a scalar multiplication $\cdot\colon \mathbb{R} \times V \to V$ such that for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$:

$$u + (v + w) = (u + v) + w, \qquad \alpha(v + w) = \alpha v + \alpha w, \tag{47}$$

$$v + w = w + v, \qquad (\alpha + \beta)v = \alpha v + \beta v, \tag{48}$$

$$\text{there is } 0 \in V \text{ with } v + 0 = v, \qquad \alpha(\beta v) = (\alpha\beta)v, \tag{49}$$

$$\text{there is } -v \in V \text{ with } v + (-v) = 0, \qquad 1 \cdot v = v. \tag{50}$$

Example 25. The spaces $\mathbb{R}^n$ and $\mathbb{R}^{m\times n}$ with entrywise addition and scalar multiplication are real vector spaces. ◊

A complex vector space is defined in the same way, with $\mathbb{C}$ in place of $\mathbb{R}$.

3.1.2. Basis and dimension. Vectors $v_1, \ldots, v_k \in V$ are linearly independent if $\sum_i \alpha_i v_i = 0$ implies $\alpha_1 = \cdots = \alpha_k = 0$. Their span is the set of all linear combinations $\sum_i \alpha_i v_i$. A basis of V is a linearly independent set that spans V. Every $v \in V$ is then a unique linear combination of the basis vectors. All bases of a finite-dimensional space have the same size, called the dimension $\dim V$. We have $\dim \mathbb{R}^n = n$ and $\dim \mathbb{R}^{m\times n} = mn$.

3.2. Vectors

3.2.1. Notation. A vector $v \in \mathbb{R}^n$ is a column with entries v_i:

$$v = (v_1, \ldots, v_n) = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}, \qquad v^T = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}. \tag{51}$$

3.2.2. Inner product. The standard inner product on $\mathbb{R}^n$ is

$$\langle v, w \rangle = v^T w = \sum_{i=1}^{n} v_i w_i. \tag{52}$$

For all $u, v, w \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$:

$$\langle v, w\rangle = \langle w, v\rangle \qquad \text{(symmetry)}, \tag{53}$$

$$\langle u, \alpha v + \beta w\rangle = \alpha\langle u, v\rangle + \beta\langle u, w\rangle \qquad \text{(linearity)}, \tag{54}$$

$$\langle v, v\rangle \geq 0, \quad \langle v, v\rangle = 0 \iff v = 0 \qquad \text{(positive definiteness)}. \tag{55}$$

The induced norm is $\|v\| = \sqrt{\langle v, v\rangle}$. Vectors v, w are orthogonal if $\langle v, w\rangle = 0$.

3.2.3. Outer product. For $v \in \mathbb{R}^m$ and $w \in \mathbb{R}^n$, the outer product is the $m \times n$ matrix

$$vw^T = (v_i w_j)_{ij}\,. \tag{56}$$

It has rank 1 if $v, w \neq 0$.

3.2.4. Canonical basis. The canonical basis vector $e_i \in \mathbb{R}^n$ has entries $(e_i)_j = \delta_{ij}$, i.e. a 1 in position i and 0 elsewhere. Then $\langle e_i, e_j\rangle = \delta_{ij}$, and every $v \in \mathbb{R}^n$ expands as

$$v = \sum_{i=1}^{n} v_i e_i, \qquad v_i = \langle e_i, v\rangle. \tag{57}$$

3.2.5. Expansion in an orthonormal basis. A set $\{h_1, \ldots, h_n\} \subset \mathbb{R}^n$ is an orthonormal basis (ONB) if $\langle h_i, h_j\rangle = \delta_{ij}$. Orthonormal vectors are linearly independent, so n of them form a basis. Hence $v = \sum_i c_i h_i$ for unique c_i, and $\langle h_j, v\rangle = \sum_i c_i \langle h_j, h_i\rangle = c_j$. Therefore

$$v = \sum_{i=1}^{n} \langle h_i, v\rangle\, h_i, \qquad \langle v, w\rangle = \sum_{i=1}^{n} \langle h_i, v\rangle\langle h_i, w\rangle, \qquad \|v\|^2 = \sum_{i=1}^{n} \langle h_i, v\rangle^2. \tag{58}$$

Example 26. For $n = 2$, take $h_\pm = (e_1 \pm e_2)/\sqrt{2}$. Then

$$v = \frac{v_1 + v_2}{\sqrt{2}}\, h_+ + \frac{v_1 - v_2}{\sqrt{2}}\, h_-. \tag{59}$$

$\Diamond$

3.3. Matrices

3.3.1. Notation. We write

$$A = (a_{ij})_{ij} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{R}^{m\times n} \tag{60}$$

for the matrix with entry a_{ij} in row i and column j. The transpose is $A^T = (a_{ji})_{ij}$. It reverses products: $(AB)^T = B^T A^T$.

3.3.2. Canonical basis. The canonical basis of $\mathbb{R}^{m\times n}$ consists of the matrices $E_{ij} = e_i e_j^T$, which have a 1 in position (i, j) and 0 elsewhere. Every $A \in \mathbb{R}^{m\times n}$ expands as

$$A = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} E_{ij}. \tag{61}$$

3.3.3. Matrix–vector multiplication. For $A \in \mathbb{R}^{m\times n}$ and $v \in \mathbb{R}^n$, the product $Av \in \mathbb{R}^m$ has entries

$$(Av)_i = \sum_{j=1}^{n} a_{ij} v_j. \tag{62}$$

Equivalently, $Av = \sum_j v_j\, Ae_j$, a linear combination of the columns Ae_j of A. In particular, $a_{ij} = e_i^T A e_j$. The rank of A is the dimension of its range $\{Av : v \in \mathbb{R}^n\}$. For an outer product,

$$(vw^T)u = \langle w, u\rangle\, v. \tag{63}$$

3.3.4. Matrix–matrix multiplication. For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, the product $AB \in \mathbb{R}^{m\times p}$ has entries

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}. \tag{64}$$

This is composition: $(AB)v = A(Bv)$. Multiplication is associative and in general not commutative.

3.3.5. Trace. For $A \in \mathbb{R}^{n\times n}$,

$$\operatorname{tr}(A) = \sum_{i=1}^{n} a_{ii} = \sum_{i=1}^{n} e_i^T A e_i. \tag{65}$$

The trace is linear, and $\operatorname{tr}(A^T) = \operatorname{tr}(A)$.

3.3.6. Cyclicity of the trace. For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$,

$$\operatorname{tr}(AB) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} b_{ji} = \sum_{j=1}^{n}\sum_{i=1}^{m} b_{ji} a_{ij} = \operatorname{tr}(BA). \tag{66}$$

Consequently, $\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB)$ whenever the products are defined.

Note: In general, $\operatorname{tr}(ABC) \neq \operatorname{tr}(ACB)$. For example, let $A = e_1 e_2^T$, $B = e_2 e_1^T$, $C = e_1 e_1^T$ in $\mathbb{R}^{2\times 2}$. Then

$$\operatorname{tr}(ABC) = \operatorname{tr}(e_1 e_2^T e_2 e_1^T e_1 e_1^T) = \langle e_2, e_2\rangle\langle e_1, e_1\rangle\langle e_1, e_1\rangle = 1, \tag{67}$$

but

$$\operatorname{tr}(ACB) = \operatorname{tr}(e_1 e_2^T e_1 e_1^T e_2 e_1^T) = \langle e_2, e_1\rangle\langle e_1, e_2\rangle\langle e_1, e_1\rangle = 0. \tag{68}$$

3.3.7. Trace inner product. The trace (Hilbert–Schmidt) inner product on $\mathbb{R}^{m\times n}$ is

$$\langle A, B\rangle = \operatorname{tr}(A^T B) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} b_{ij}. \tag{69}$$

It has the same properties as the standard inner product on vectors:

$$\langle A, B\rangle = \langle B, A\rangle \qquad \text{(symmetry)}, \tag{70}$$

$$\langle C, \alpha A + \beta B\rangle = \alpha\langle C, A\rangle + \beta\langle C, B\rangle \qquad \text{(linearity)}, \tag{71}$$

$$\langle A, A\rangle \geq 0, \quad \langle A, A\rangle = 0 \iff A = 0 \qquad \text{(positive definiteness)}. \tag{72}$$

The induced norm is the Frobenius norm $\|A\|_F = \sqrt{\langle A, A\rangle} = \left(\sum_{i,j} a_{ij}^2\right)^{1/2}$.

Remark 27. *Let* $\operatorname{vec}(A) \in \mathbb{R}^{mn}$ *be the vector obtained by stacking the columns of* A*. The trace inner product is the standard inner product of these vectors:*

$$\langle A, B\rangle = \operatorname{tr}(A^T B) = \langle \operatorname{vec}(A), \operatorname{vec}(B)\rangle. \tag{73}$$

3.3.8. Bilinear forms as traces. For $A \in \mathbb{R}^{m\times n}$, $v \in \mathbb{R}^n$ and $w \in \mathbb{R}^m$,

$$w^T A v = \operatorname{tr}(A v w^T). \tag{74}$$

Proof. Using $(vw^T)_{ji} = v_j w_i$,

$$w^T A v = \sum_{i,j} w_i a_{ij} v_j = \sum_{i,j} a_{ij} (vw^T)_{ji} = \sum_{i} (Avw^T)_{ii} = \operatorname{tr}(Avw^T). \quad \square \tag{75}$$

In terms of the two inner products,

$$\langle w, Av\rangle = \langle wv^T, A\rangle. \tag{76}$$

3.3.9. Projectors: trace equals rank. A matrix $P \in \mathbb{R}^{n\times n}$ is an orthogonal projector if $P^2 = P = P^T$. Let U be its range, let $r = \operatorname{rank}(P)$, and let $u_1, \dots, u_r$ be an ONB of U. Then

$$P = \sum_{i=1}^{r} u_i u_i^T, \qquad \operatorname{tr}(P) = \sum_{i=1}^{r} \|u_i\|^2 = r = \operatorname{rank}(P). \tag{77}$$

Proof. Let $Q = \sum_i u_i u_i^T$. If $u = Pw \in U$ for some w, then $Pu = P^2 w = u$, and $Qu = u$ by the ONB expansion. If $x \perp U$, then $\langle Px, y\rangle = \langle x, Py\rangle = 0$ for all y, so $Px = 0$, and also $Qx = 0$. Since $\mathbb{R}^n = U \oplus U^\perp$, we get $P = Q$. The trace formula follows from $\operatorname{tr}(u_i u_i^T) = \langle u_i, u_i\rangle$. □

So on projectors, the rank is the linear function tr. The extension to positive semidefinite matrices is in Section 3.5.

Example 28. The projector $h_+ h_+^T = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has trace and rank 1. ◊

3.3.10. Expansion of a matrix in an ONB. Let $\{B_k\}_{k=1}^{mn}$ be an ONB of $\mathbb{R}^{m\times n}$ with respect to the trace inner product, i.e. $\operatorname{tr}(B_k^T B_l) = \delta_{kl}$. Write $A = \sum_k c_k B_k$. As for vectors, taking the inner product with B_l gives $\langle B_l, A\rangle = \sum_k c_k \langle B_l, B_k\rangle = c_l$. Therefore

$$A = \sum_{k=1}^{mn} \langle B_k, A\rangle\, B_k = \sum_{k=1}^{mn} \operatorname{tr}(B_k^T A)\, B_k. \tag{78}$$

Moreover,

$$\|A\|_F^2 = \sum_{k=1}^{mn} \operatorname{tr}(B_k^T A)^2. \tag{79}$$

Canonical basis. The matrices $e_i e_j^T$ form an ONB, and

$$\langle e_i e_j^T, A\rangle = \operatorname{tr}(e_j e_i^T A) = e_i^T A e_j = a_{ij}. \tag{80}$$

Bases induced by an ONB of $\mathbb{R}^n$. If $\{h_i\}$ is an ONB of $\mathbb{R}^n$, then $\{h_i h_j^T\}$ is an ONB of $\mathbb{R}^{n\times n}$, since

$$\langle h_i h_j^T, h_k h_l^T\rangle = \operatorname{tr}(h_j h_i^T h_k h_l^T) = \langle h_i, h_k\rangle\langle h_l, h_j\rangle = \delta_{ik}\delta_{jl}. \tag{81}$$

With $\langle h_i h_j^T, A\rangle = h_i^T A h_j$, this gives

$$A = \sum_{i,j} (h_i^T A h_j)\, h_i h_j^T, \qquad I = \sum_i h_i h_i^T. \tag{82}$$

Example 29 (Pauli matrices and qubits)**.** In this example only, we work over $\mathbb{C}$. The adjoint is $A^\dagger = \overline{A}^T$, and the trace inner product on $\mathbb{C}^{m\times n}$ is

$$\langle A, B\rangle = \operatorname{tr}(A^\dagger B) = \sum_{i=1}^{m}\sum_{j=1}^{n} \overline{a_{ij}}\, b_{ij}. \tag{83}$$

It is conjugate symmetric, $\langle A, B\rangle = \overline{\langle B, A\rangle}$, linear in the second argument, antilinear in the first, and positive definite. The Pauli matrices are

$$\begin{aligned} \sigma_0 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \sigma_1 = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \\ \sigma_2 = Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{aligned} \tag{84}$$

Each σ_a is Hermitian, i.e. $\sigma_a^\dagger = \sigma_a$, and satisfies $\sigma_a^2 = I$. Moreover, $XY = iZ$, $YZ = iX$, $ZX = iY$, and distinct X, Y, Z anticommute. Thus $\sigma_a\sigma_b$ is a phase times a traceless Pauli matrix for $a \neq b$, so

$$\langle \sigma_a, \sigma_b\rangle = \operatorname{tr}(\sigma_a\sigma_b) = 2\delta_{ab}. \tag{85}$$

Hence $\{\sigma_a/\sqrt{2}\}_{a=0}^{3}$ is an ONB of $\mathbb{C}^{2\times 2}$, and every $A \in \mathbb{C}^{2\times 2}$ expands as

$$A = \frac{1}{2}\sum_{a=0}^{3} \operatorname{tr}(\sigma_a A)\,\sigma_a. \tag{86}$$

For a general matrix,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{a+d}{2}\,I + \frac{b+c}{2}\,X + \frac{i(b-c)}{2}\,Y + \frac{a-d}{2}\,Z. \tag{87}$$

For example,

$$e_1 e_1^T = \frac{I+Z}{2}, \qquad e_1 e_2^T = \frac{X+iY}{2}, \qquad \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{X+Z}{\sqrt{2}}. \tag{88}$$

If $A = A^\dagger$, all coefficients $\operatorname{tr}(\sigma_a A)$ are real. In particular, a qubit state is a density matrix $\rho \in \mathbb{C}^{2\times 2}$, i.e. $\rho = \rho^\dagger$ is positive semidefinite (Section 3.5) with $\operatorname{tr}(\rho) = 1$. The expansion gives

$$\rho = \frac{1}{2}\left(I + r_1 X + r_2 Y + r_3 Z\right), \qquad r_a = \operatorname{tr}(\sigma_a \rho) \in \mathbb{R}, \tag{89}$$

where the coefficient of I is fixed by $\operatorname{tr}(\rho) = 1$.

The matrix $J = iY = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is real, with $J^T = -J$ and $J^2 = -I$. Thus $\{I, X, J, Z\}/\sqrt{2}$ is an ONB of the real space $\mathbb{R}^{2\times 2}$. ◊

3.4. Eigenvalues and eigenvectors

3.4.1. Definition. Let $A \in \mathbb{R}^{n\times n}$. A vector $v \neq 0$ is an eigenvector of A with eigenvalue λ if

$$Av = \lambda v. \tag{90}$$

So A acts on the line $\mathbb{R}v$ as multiplication by λ. A number λ is an eigenvalue if and only if $A - \lambda I$ is not invertible, i.e. $\det(A - \lambda I) = 0$. A real matrix need not have real eigenvalues: J is a rotation by 90°, preserves no line, and has eigenvalues $\pm i$.

3.4.2. Geometric picture. Let $D = \operatorname{diag}(2, \frac{1}{2})$. It maps the unit circle $\{x : \|x\| = 1\}$ onto the ellipse

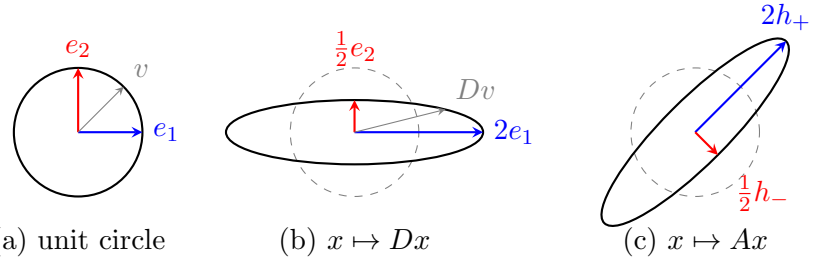
$$\left\{ y \in \mathbb{R}^2 : \frac{y_1^2}{4} + 4y_2^2 = 1 \right\}. \tag{91}$$

The semi-axes of this ellipse lie along e_1, e_2, and their lengths are the eigenvalues $2, \frac{1}{2}$. The eigenvectors are exactly the directions that D only stretches. Any other vector also changes direction, e.g. $v = (1,1)/\sqrt{2}$ maps to $Dv = (2, \frac{1}{2})/\sqrt{2}$.

For a symmetric matrix that is not diagonal, the ellipse is tilted, and its axes are the eigenvectors. For example, let

$$A = \frac{1}{4}\begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}, \qquad Ah_+ = 2h_+, \qquad Ah_- = \frac{1}{2}h_-, \tag{92}$$

with $h_\pm = (e_1 \pm e_2)/\sqrt{2}$. In the ONB $\{h_+, h_-\}$, the matrix A acts as D. Diagonalizing A means choosing the ONB in which the ellipse is axis-aligned. See Figure 1.



(a) unit circle (b) $x \mapsto Dx$ (c) $x \mapsto Ax$

FIGURE 1. Image of the unit circle under a diagonal and a non-diagonal symmetric matrix. The axes of the ellipse are the eigenvectors scaled by the eigenvalues.

3.4.3. Spectral theorem. A matrix $U \in \mathbb{R}^{n\times n}$ is orthogonal if $U^TU = I$, i.e. if its columns form an ONB. Every symmetric matrix $A \in \mathbb{R}^{n\times n}$ has real eigenvalues $\lambda_1, \dots, \lambda_n$ and an ONB of eigenvectors $u_1, \dots, u_n$, so

$$A = U\Lambda U^T = \sum_{i=1}^{n} \lambda_i u_i u_i^T, \qquad U = \begin{pmatrix} | & & | \\ u_1 & \cdots & u_n \\ | & & | \end{pmatrix}, \quad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n). \tag{93}$$

Eigenvectors with different eigenvalues are automatically orthogonal: if $Au = \lambda u$ and $Av = \mu v$, then $\lambda\langle u, v\rangle = \langle Au, v\rangle = \langle u, Av\rangle = \mu\langle u, v\rangle$. The image of the unit sphere is an ellipsoid with semi-axes $\lambda_i u_i$, which is flat if some $\lambda_i = 0$.

3.4.4. Consequences. For symmetric A, cyclicity gives $\operatorname{tr}(A) = \operatorname{tr}(U^TU\Lambda) = \sum_i \lambda_i$, and $\operatorname{rank}(A)$ is the number of nonzero λ_i. Moreover, $x^TAx = \sum_i \lambda_i \langle u_i, x\rangle^2$, so

$$\lambda_{\max}(A) = \max_{\|x\|=1} x^TAx, \qquad \lambda_{\min}(A) = \min_{\|x\|=1} x^TAx. \tag{94}$$

A symmetric matrix is a projector if and only if all its eigenvalues lie in $\{0, 1\}$, since $P^2 = P$ means $\lambda^2 = \lambda$.

3.5. Positive semidefinite matrices

Let $\mathcal{S}^n = \{A \in \mathbb{R}^{n\times n} : A = A^T\}$, a vector space of dimension $n(n+1)/2$ with inner product $\langle A, B\rangle = \operatorname{tr}(AB)$.

3.5.1. Definition. A matrix $A \in \mathcal{S}^n$ is positive semidefinite (PSD), written $A \succeq 0$, if $x^TAx \geq 0$ for all $x \in \mathbb{R}^n$. We write $\mathcal{S}^n_+$ for the set of PSD matrices. Symmetry is part of the definition: $\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ satisfies $x^TAx = \|x\|^2$ but is not PSD.

3.5.2. Theorem (equivalent definitions). For $A \in \mathcal{S}^n$, the following are equivalent:

(a) Quadratic form: $x^TAx \geq 0$ for all $x \in \mathbb{R}^n$.
(b) Eigenvalues: all eigenvalues of A are non-negative.
(c) Sum of rank-one terms: $A = \sum_{l=1}^{m} x_l x_l^T$ for some $x_l \in \mathbb{R}^n$.
(d) Hermitian square: $A = B^TB$ for some $B \in \mathbb{R}^{k\times n}$.
(e) Gram matrix: $a_{ij} = \langle v_i, v_j\rangle$ for some vectors $v_1, \dots, v_n \in \mathbb{R}^k$.
(f) Cholesky factorization: $A = LL^T$ for some lower triangular $L \in \mathbb{R}^{n\times n}$.
(g) Self-duality: $\operatorname{tr}(AB) \geq 0$ for all $B \in \mathcal{S}^n_+$.

Proof. (a) $\Rightarrow$ (b): If $Au = \lambda u$ with $\|u\| = 1$, then $\lambda = u^TAu \geq 0$.
(b) $\Rightarrow$ (c): Take $x_i = \sqrt{\lambda_i}\, u_i$ in the spectral decomposition.
(c) $\Rightarrow$ (d): Let B have rows x_l^T. Then $B^TB = \sum_l x_l x_l^T$.
(d) $\Rightarrow$ (e): Let v_j be the columns of B. Then $(B^TB)_{ij} = \langle v_i, v_j\rangle$.
(e) $\Rightarrow$ (a): $x^TAx = \sum_{i,j} x_i x_j \langle v_i, v_j\rangle = \|\sum_i x_i v_i\|^2 \geq 0$.

(b) ⇒ (f): The matrix $B = \Lambda^{1/2}U^T$ is square with $B^TB = A$. Take a QR decomposition $B = QR$, with Q orthogonal and R upper triangular. Then $A = R^TQ^TQR = R^TR$, so $L = R^T$ works.

(f) ⇒ (d): Take $B = L^T$.

(a) ⇒ (g): By (c), any $B \in \mathcal{S}^n_+$ can be written as $B = \sum_l x_l x_l^T$. Then $\operatorname{tr}(AB) = \sum_l x_l^T A x_l \geq 0$.

(g) ⇒ (a): The matrix xx^T is PSD by (c), and $\operatorname{tr}(Axx^T) = x^TAx$. □

3.5.3. Remarks. The matrix A is positive definite, written $A \succ 0$, if $x^TAx > 0$ for all $x \neq 0$. This is equivalent to all eigenvalues being positive, and to B in (d) having full column rank.

Example 30. The matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ is PSD, with eigenvalues $2, 0$. The matrix $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ is not, since $x = (1, -1)$ gives $x^TAx = -2$. ♢

Trace as a proxy for rank. Let $A \succeq 0$ with $A \neq 0$. Since $0 \leq \lambda_i \leq \lambda_{\max}(A)$,

$$\frac{\operatorname{tr}(A)}{\lambda_{\max}(A)} \leq \operatorname{rank}(A), \tag{95}$$

with equality if and only if A is a multiple of a projector. If all eigenvalues of A are at most 1, this gives $\operatorname{tr}(A) \leq \operatorname{rank}(A)$. This is why the rank is often replaced by the trace in semidefinite relaxations.

Example 31 (Qubits)**.** Over $\mathbb{C}$, a matrix is PSD if $A = A^\dagger$ and $x^\dagger Ax \geq 0$ for all $x \in \mathbb{C}^n$. The matrix $r_1X + r_2Y + r_3Z$ is traceless and squares to $\|r\|^2 I$, so it has eigenvalues $\pm\|r\|$. Thus ρ has eigenvalues $(1 \pm \|r\|)/2$, and $\rho \succeq 0$ if and only if $\|r\| \leq 1$. Qubit states form the Bloch ball. ♢

3.6. Convex sets and cones

3.6.1. Löwner order. For $A, B \in \mathcal{S}^n$, we write $A \succeq B$ if $A - B \succeq 0$. This is a partial order: it is reflexive, transitive, and antisymmetric, since $A \succeq B \succeq A$ gives $\lambda_i(A - B) = 0$ for all i, hence $A = B$. It is not a total order: there are matrices A, B that are incomparable, meaning that neither $A \succeq B$ nor $B \succeq A$ holds. Can you find two incomparable 2×2 matrices?

Cones, convex sets, and convex hulls are defined in Section 1.2.

3.6.2. The PSD cone. The set $\mathcal{S}^n_+$ is a convex cone: if $A, B \succeq 0$ and $\alpha, \beta \geq 0$, then

$$x^T(\alpha A + \beta B)x = \alpha\, x^TAx + \beta\, x^TBx \geq 0. \tag{96}$$

It is closed, since it is an intersection of the closed half-spaces $\{A : \langle xx^T, A\rangle \geq 0\}$. By (c), it is the set of finite sums of the rank-one matrices xx^T. The set of PSD matrices of rank at most 1 is a cone, but it is not convex for $n \geq 2$. The set of PSD matrices of trace 1 is the convex hull of the rank-one projectors:

$$\{A \in \mathcal{S}^n_+ : \operatorname{tr}(A) = 1\} = \operatorname{conv}\{xx^T : \|x\| = 1\}. \tag{97}$$

Here "⊇" holds since the left-hand side is convex and contains each xx^T. For "⊆", write $A = \sum_i \lambda_i u_i u_i^T$ with $\lambda_i \geq 0$ and $\sum_i \lambda_i = \operatorname{tr}(A) = 1$. This set is the real analogue of the set of density matrices.

For $n = 2$, the matrix $A = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$ has eigenvalues $\frac{1}{2}\big(a + c \pm \sqrt{(a-c)^2 + 4b^2}\big)$. Thus $A \succeq 0$ if and only if $a + c \geq \sqrt{(a-c)^2 + 4b^2}$. In the orthonormal coordinates $(a + c, a - c, 2b)/\sqrt{2}$, this is a circular cone in $\mathbb{R}^3$. Its boundary consists of the matrices of rank at most 1. See Figure 2(b).

3.6.3. Dual cone. Let V be a vector space with an inner product, and let $K \subseteq V$ be a cone. The dual cone is

$$K^* = \{y \in V : \langle y, x\rangle \geq 0 \text{ for all } x \in K\}. \tag{98}$$

It is always a closed convex cone, as an intersection of closed half-spaces. If K is a closed convex cone, then $K^{**} = K$. A cone is self-dual if $K^* = K$.

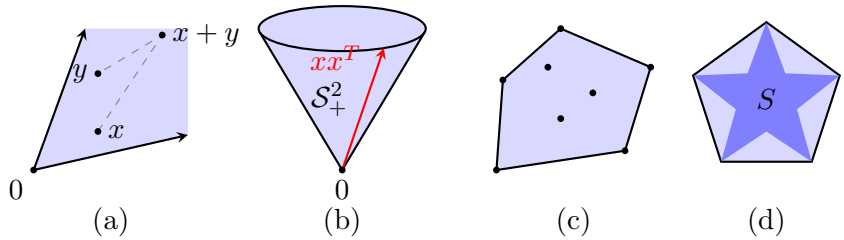


FIGURE 2. (a) A convex cone, which contains $x + y$ for all its points x, y. (b) The cone $\mathcal{S}^2_+$, with a boundary ray generated by a rank-one matrix xx^T. (c) The convex hull of a finite set of points. (d) The convex hull (light) of a non-convex set S (dark).

Example 32. The orthant $\mathbb{R}^n_+$ is self-dual: if $\langle y, x\rangle \geq 0$ for all $x \geq 0$, then $y_i = \langle y, e_i\rangle \geq 0$, and the converse is clear. The ray $K = \{\alpha e_1 : \alpha \geq 0\} \subset \mathbb{R}^2$ is not self-dual: K^* is the half-plane $\{y : y_1 \geq 0\}$. ♢

3.6.4. Self-duality of the PSD cone. In $\mathcal{S}^n$ with the trace inner product,

$$(\mathcal{S}^n_+)^* = \mathcal{S}^n_+. \tag{99}$$

This is the equivalence (a) ⇔ (g). The dual cone depends on the ambient space: in $\mathbb{R}^{n\times n}$, the dual of $\mathcal{S}^n_+$ is $\{A : A + A^T \succeq 0\}$, which contains all antisymmetric matrices.

As a consequence, $A \succeq B$ if and only if $\operatorname{tr}(AC) \geq \operatorname{tr}(BC)$ for all $C \succeq 0$.

CHAPTER 4

Reading list

i) L. Lovász, *Semidefinite programs and combinatorial optimization*, lecture notes.
https://sites.math.washington.edu/~thomas/teaching/m514_web/Lovasz_semidef.pdf

ii) A. Gupta, R. O'Donnell, *Linear and semidefinite programming*, lecture notes, CMU course 15-859, Fall 2011. Uses the same convention for primal and dual as these notes.
https://www.cs.cmu.edu/afs/cs.cmu.edu/academic/class/15859-f11/www/notes/lpsdp.pdf

iii) J. Matoušek, B. Gärtner, *Understanding and Using Linear Programming*, Springer, 2007.
https://link.springer.com/content/pdf/10.1007/978-3-540-30717-4.pdf

iv) H. Yang, *Semidefinite Optimization and Relaxation*, lecture notes. Uses a different convention for primal and dual than these notes.
https://hankyang.seas.harvard.edu/Semidefinite/

v) G. Blekherman, P. A. Parrilo, R. R. Thomas (eds.), *Semidefinite Optimization and Convex Algebraic Geometry*, SIAM, 2013.
https://www.mit.edu/~parrilo/sdocag/

vi) *JuMP*, a modeling language for mathematical optimization in Julia.
https://jump.dev/

APPENDIX A

Proof of Farkas Lemma

The aim of this appendix is to prove the direction of Theorem 3 that is not proven in Section 1.3: if $Ax \le b$ has no solution, then there is $\lambda \ge 0$ with $A^T\lambda = \mathbf{0}$ and $b^T\lambda = -1$. We use Fourier–Motzkin elimination, following [17, Ch. 1]; see also [15]. The idea is to remove the variables $x_1, \dots, x_n$ one after another by taking non-negative combinations of the inequalities.

Lemma 33 (Fourier–Motzkin elimination step)**.** *Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and $k \in \{1, \dots, n\}$. Suppose that the columns $1, \dots, k-1$ of A are zero. Then there exist $m' \ge 0$ and a matrix $M \in \mathbb{R}^{m'\times m}$ with $M \ge 0$ such that*

(a) the columns $1, \dots, k$ of $A' := MA$ are zero, and

(b) with $b' := Mb$, the system $Ax \le b$ has a solution if and only if the system $A'x \le b'$ has a solution.

(If $m' = 0$, the system $A'x \le b'$ is empty, and every $x \in \mathbb{R}^n$ is a solution.)

Proof. Denote the entries of A by a_{ij}. Split the row indices into

$$P = \{i : a_{ik} > 0\}\,, \quad N = \{i : a_{ik} < 0\}\,, \quad Z = \{i : a_{ik} = 0\}\,. \tag{100}$$

The matrix M has one row for each $i \in Z$ and one row for each pair $(i,j) \in P \times N$:

$$\text{row } i \in Z\text{: } e_i^T\,, \qquad \text{row } (i,j) \in P\times N\text{: } \frac{1}{a_{ik}}e_i^T + \frac{1}{|a_{jk}|}e_j^T\,. \tag{101}$$

All entries of M are non-negative.

(a) Row $i \in Z$ of MA is a_i^T, whose k-th entry is $a_{ik} = 0$. Row (i,j) of MA is $a_i^T/a_{ik} + a_j^T/|a_{jk}|$, whose k-th entry is $1 - 1 = 0$. The entries $1, \dots, k-1$ of these rows are zero, because those of a_i and a_j are.

(b) If $Ax \le b$, then $MAx \le Mb$ by (2), applied to each row of M. So every solution of $Ax \le b$ solves $A'x \le b'$.

Conversely, let x solve $A'x \le b'$. We keep all entries of x except x_k, and choose a new value t for x_k. Let $x(t)$ be the vector x with its k-th entry replaced by t. Since the k-th column of A' is zero, $x(t)$ solves $A'x(t) \le b'$ for every $t \in \mathbb{R}$. Write $a_i^T x(t) = a_{ik}t + r_i$, where $r_i = a_i^T x - a_{ik}x_k$ does not depend on t. The i-th inequality of $Ax(t) \le b$ reads

$$\begin{aligned} i \in P\text{:}\quad & t \le u_i := \frac{b_i - r_i}{a_{ik}}\,, \\ i \in N\text{:}\quad & t \ge l_i := \frac{b_i - r_i}{a_{ik}}\,, \\ i \in Z\text{:}\quad & r_i \le b_i\,. \end{aligned} \tag{102}$$

For $i \in Z$, the inequality $r_i \le b_i$ is row i of $A'x \le b'$, so it holds. For $(i,j) \in P \times N$, row (i,j) of $A'x(t) \le b'$ reads

$$\frac{a_{ik}t + r_i}{a_{ik}} + \frac{a_{jk}t + r_j}{|a_{jk}|} \le \frac{b_i}{a_{ik}} + \frac{b_j}{|a_{jk}|}\,. \tag{103}$$

The terms with t cancel, and using $|a_{jk}| = -a_{jk}$ this is equivalent to $l_j \le u_i$. Thus $l_j \le u_i$ for all $i \in P$ and $j \in N$, and therefore

$$\max_{j\in N} l_j \le \min_{i\in P} u_i\,, \tag{104}$$

where the maximum over the empty set is $-\infty$ and the minimum over the empty set is $+\infty$. Choose t between these two values. Then $x(t)$ satisfies all inequalities of $Ax \le b$. □

PROOF OF THEOREM 3: IF I) FAILS, THEN II) HOLDS. Suppose that $Ax \le b$ has no solution. Apply Lemma 33 with $k = 1$ to (A, b), then with $k = 2$ to the resulting system, and so on up to $k = n$. This gives matrices $M_1, \dots, M_n$ with non-negative entries. Their product $M = M_n \cdots M_1$ also has non-negative entries. After the last step all columns are zero, so $MA = 0$. Let $b^{(n)} = Mb$. By part (b) of Lemma 33, applied in each step, the system $0 \cdot x \le b^{(n)}$ has no solution. This system consists of the inequalities $0 \le b^{(n)}_\ell$. It has no solution, so it is not empty and $b^{(n)}_\ell < 0$ for some row ℓ. Let μ^T be row ℓ of M. Then $\mu \ge 0$, $A^T\mu = (\mu^T A)^T = \mathbf{0}$ (row ℓ of $MA = 0$), and $b^T\mu = b^{(n)}_\ell < 0$. The vector $\lambda = \mu/|b^T\mu|$ satisfies (3). □

Example 34. Consider the system

$$\begin{aligned} &\text{(i)} && x_1 + x_2 \le 1 \\ &\text{(ii)} && -x_1 \le 0 \\ &\text{(iii)} && -x_2 \le -2\,. \end{aligned}$$

It has no solution: (ii) and (iii) say $x_1 \ge 0$ and $x_2 \ge 2$, so $x_1 + x_2 \ge 2$. Eliminate x_1: here $P = \{\text{i}\}$, $N = \{\text{ii}\}$, $Z = \{\text{iii}\}$. The new system is

$$\begin{aligned} x_2 &\le 1 && \text{(i)+(ii)} \\ -x_2 &\le -2 && \text{(iii)}\,. \end{aligned} \tag{105}$$

Eliminate x_2: the sum of the two inequalities is $0 \le -1$. It is the sum of (i), (ii), (iii). So $\lambda = (1, 1, 1)^T$, and indeed $A^T\lambda = (1 - 1 + 0, 1 + 0 - 1)^T = \mathbf{0}$ and $b^T\lambda = 1 + 0 - 2 = -1$. ◊

Remark 35. *Fourier–Motzkin elimination is also an algorithm to decide whether $Ax \le b$ has a solution. It is not efficient: one step can replace m inequalities by up to $|P| \cdot |N| \le m^2/4$ inequalities.*

To summarize this appendix: eliminating all variables by non-negative combinations of the inequalities turns an infeasible system $Ax \le b$ into an inequality $0 \le b^T\mu$ with $b^T\mu < 0$, and the weights μ, rescaled, are the vector λ of Farkas Lemma.

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